

Overview

The Qualcomm Memory Dump Collector (QMDC) is a tool for collecting memory dump files from Qualcomm-based devices.

Open Source Repository

<https://github.com/qualcomm/qcom-memory-dump-collector>

Distributed Components

QMDC is a zipped package with list of below component bundled:

- QMDC executable
- QMDC User Guide
- Qualcomm User Space Driver (install optional)

Supported Operating Systems

OS	Description
Linux x64	Ubuntu 24.04 and newer
Windows x64	Windows 11
Windows ARM64	Windows 11
WSL	Ubuntu 24.04 and newer

Setup

1. Download the zip package for your platform from [Qualcomm Software Center \(QSC\)](#). – Search for “Qualcomm Memory Dump Collector”
2. Unzip the downloaded zip package.
3. Driver installation(optional)- Installing Qualcomm user-space drivers is optional if the Qualcomm kernel drivers are already present. If drivers have not been installed, the user-space driver included in this package may be used as an alternative. Driver installation is a one-time process; please refer to the "README" provided with the accompanying user-space driver zip bundle for detailed installation instructions.

Usage

Prerequisites

Before using QMDC, ensure the device is in Crash mode:

Put the device in crash mode before collecting memory dump files

The following commands can be used to set a device to crash mode:

```
adb root
adb wait-for-device
adb remount
```

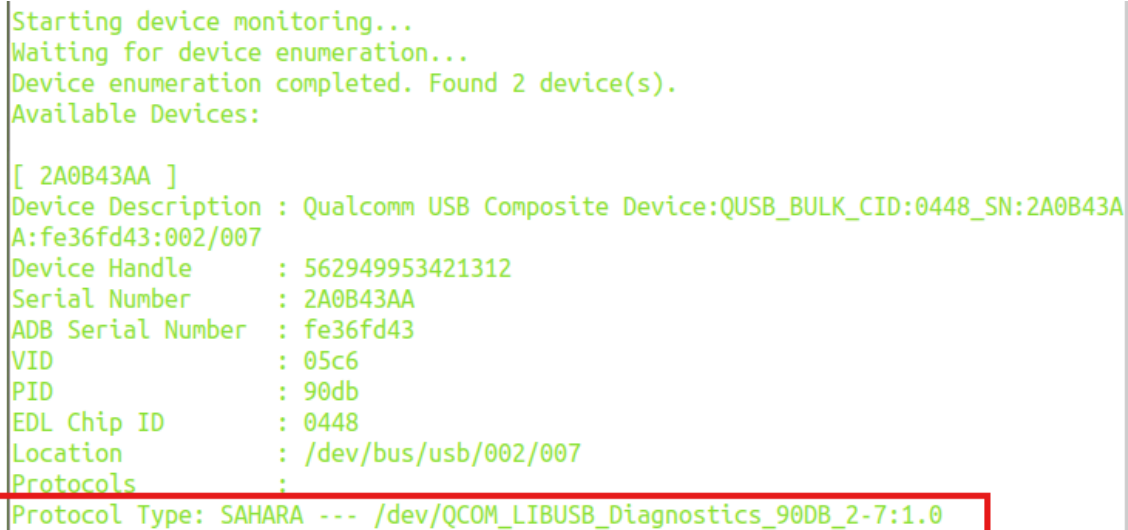
```
adb wait-for-device
adb shell "echo c > /proc/sysrq-trigger"
```

Crash Mode in Linux Verification

If the device is in Crash mode, the output will include a line similar to:

```
Protocol Type: SAHARA --- /dev/QCOM_LIBUSB_Diagnostics_90DB_2-7:1.0
```

(See the screenshot below for an example.)



The screenshot displays the output of an ADB session. It begins with 'Starting device monitoring...' and 'Waiting for device enumeration...'. After 'Device enumeration completed. Found 2 device(s).', it lists 'Available Devices:'. One device is shown with details: '[2A0B43AA]', 'Device Description : Qualcomm USB Composite Device:QUSB_BULK_CID:0448_SN:2A0B43AA:fe36fd43:002/007', 'Device Handle : 562949953421312', 'Serial Number : 2A0B43AA', 'ADB Serial Number : fe36fd43', 'VID : 05c6', 'PID : 90db', 'EDL Chip ID : 0448', 'Location : /dev/bus/usb/002/007', and 'Protocols :'. The final line, 'Protocol Type: SAHARA --- /dev/QCOM_LIBUSB_Diagnostics_90DB_2-7:1.0', is highlighted with a red rectangular box.

```
Starting device monitoring...
Waiting for device enumeration...
Device enumeration completed. Found 2 device(s).
Available Devices:

[ 2A0B43AA ]
Device Description : Qualcomm USB Composite Device:QUSB_BULK_CID:0448_SN:2A0B43AA:fe36fd43:002/007
Device Handle      : 562949953421312
Serial Number      : 2A0B43AA
ADB Serial Number   : fe36fd43
VID                : 05c6
PID                : 90db
EDL Chip ID        : 0448
Location           : /dev/bus/usb/002/007
Protocols          :
Protocol Type: SAHARA --- /dev/QCOM_LIBUSB_Diagnostics_90DB_2-7:1.0
```

Figure 1 - Successfully set device to Crash mode (Linux)

Crash Mode in Windows Verification

If the device is in Crash mode, the output will include a line similar to:

```
Protocol Type: SAHARA --- Qualcomm HS-USB Diagnostics 90DB
```

(See the screenshot below for an example.)

```
Starting device monitoring...
Waiting for device enumeration...
Device enumeration completed. Found 1 device(s).
Available Devices:

[ 1E4EF136 ]
Device Description : Qualcomm USB Composite Device 90DB (0011)
Device Handle      : 281474976710656
Serial Number      : 1E4EF136
ADB Serial Number   : 21c0df74
VID                : 05C6
PID                : 90DB
EDL Chip ID        : UNKNOWN
Location           : 0000.0000.0000.003.000.000.000.000
Protocol Type: SAHARA --- Qualcomm HS-USB Diagnostics 90DB

Total devices found: 1
Stopping device monitoring...
```

Figure 2 - Successfully set device to Crash mode (Windows)

Crash Mode in WSL Verification

If the device is in Crash mode, the output will include a line similar to:

```
Protocol Type: SAHARA --- /dev/QCOM_LIBUSB_Diagnostics_90DB_2-7:1.0
```

(See the screenshot below for an example.)

```
Starting device monitoring...
Waiting for device enumeration...
Device enumeration completed. Found 1 device(s).
Available Devices:

[ 2A0B43AA ]
Device Description : Qualcomm USB Composite Device:QUSB_BULK_CID:0448_SN:2A0B43AA:fe36fd43:002/007
Device Handle      : 281474976710656
Serial Number      : 2A0B43AA
ADB Serial Number   : fe36fd43
VID                : 05c6
PID                : 90db
EDL Chip ID        : 0448
Location           : /dev/bus/usb/002/007
Protocol Type: SAHARA --- /dev/QCOM_LIBUSB_Diagnostics_90DB_2-7:1.0
```

Figure 3 - Successfully set device to Crash mode (WSL)

Run QMDC

Below is a commonly used command within QMDC; for a comprehensive list, refer to the [QMDC Commands section](#).

Collect a crash dump from a device:

```
./qmdc --crash-collection --device=<DEVICE_ID> --path-name=/path/to/store/files
```

Supported Features

QMDC Commands

To view a list of all available commands, run `./qmdc --help` or `./qmdc -h` from the command line. The following table lists all supported QMDC commands, along with a brief description of each command and its required and optional arguments. See the [QMDC Arguments section](#) for more information about the arguments.

Command	Required Arguments	Optional Arguments	Description
<code>--crash-collection</code>	<code>--device</code> <code>--path-name</code>	<code>--verbose</code>	Download memory dumps from a device.
<code>--help, -h</code>			Display help information.
<code>--devices</code>		<code>--verbose</code>	List all available device identifiers.
<code>--version</code>			Display QMDC application version.

QMDC Arguments

The following table lists all available arguments, their expected values, and descriptions.

Option	Value	Description
<code>--device</code>	<ID>	Specify target device identifier for device operation. Use SERIAL NUMBER OR ADB Serial Number Or Device Description from <code>--devices</code> command.
<code>--path-name</code>	<DIR_PATH>	Absolute path for directory to store memory dump files.
<code>--verbose</code>		Enable verbose logging output. Shows detailed operation logs and debug information.

Logs

Debug logs capture detailed information about QMDC operations and can assist in troubleshooting. They can be found in the following location:

Linux

`/var/tmp/QFS/QMDC/Logs/`

Windows

`C:\ProgramData\QFS\QMDC\Logs`

WSL

`/var/tmp/QFS/QMDC/Logs/`

Limitations

1. ADB Cannot Run Simultaneously with QMDC

QMDC and ADB cannot access the same USB device simultaneously. Only one application can hold a handle to a USB device interface at a time. If ADB is running, stop it before using QMDC:

```
adb kill-server
```

2. Crash Mode Required

All QMDC device operations require the device to be in Crash mode. If the device is not in Crash mode, commands such as `./qmdc --devices` will not detect it. See the [Crash mode](#) section for setup instructions.

3. Single Device at a Time

QMDC supports collecting memory dumps from only one device at a time. Multi-device parallel operations are not supported.

FAQ

Below are some common issues and their solutions.

1. QMDC reports that the device has not been found.

Please ensure Qualcomm Drivers have been installed. See the [Setup](#) for details.